BABU BANARASI DAS UNIVERSITY

2026-2027



SCHOOL OF COMPUTER APPLICATIONS

No SQL Project On

Student Management System Using MongoDB

Submitted to:
Mr. Harendra Singh
Department: BCA (DS & AI)
Date: 25th September

Submitted by:
Harleen Kaur
Roll no: (44)
University roll no
1250258209

OBJECTIVES

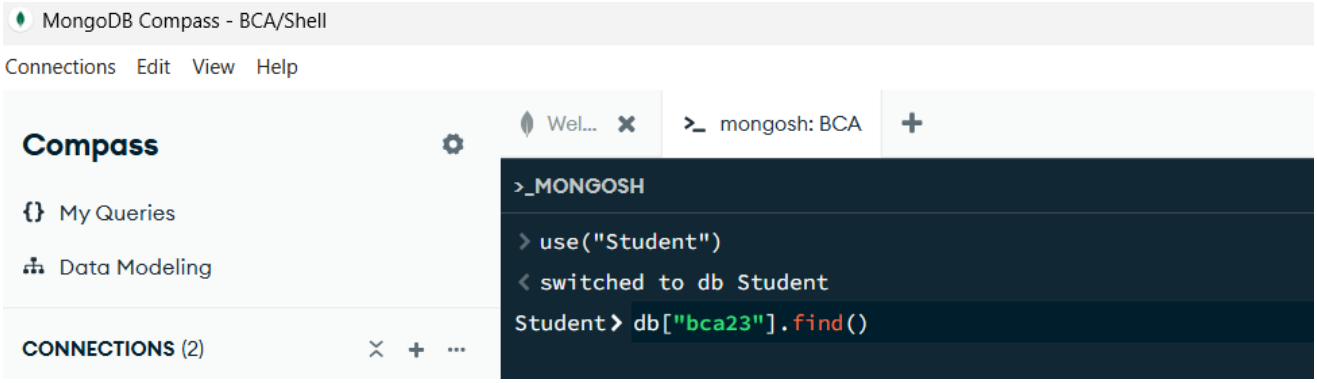
- To understand the concept of NoSQL databases.
- To learn the basic features of MongoDB.
- To create a database and collection.
- To insert 50 student records.
- To perform CRUD operations.
- To use comparison operators.
- To perform sorting and filtering.
- To understand AND and OR operations.

1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.

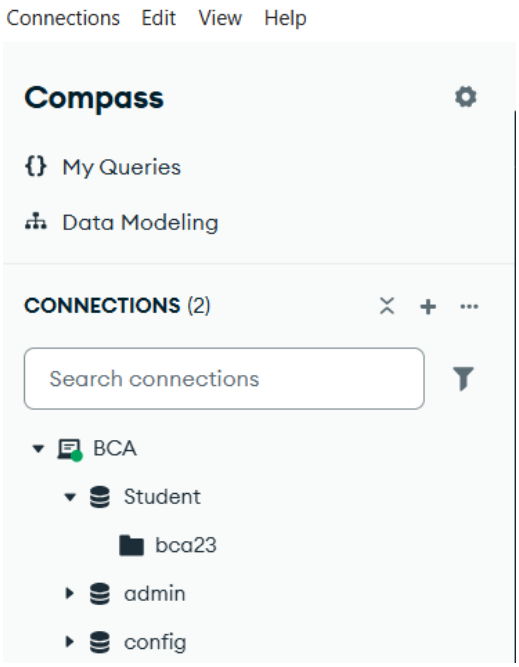


2. CREATE COLLECTION

MongoDB Query:

db. create Collection("students")

Explanation: Creates a collection named students.



3. INSERT 50 STUDENTS

MongoDB Query:

```
db.bca23.insertMany([
 {
   student_id: 1,
   name: "Aarav Sharma",
   age: 20,
   gender: "Male",
   semester: 1,
   section: "A",
   email: "aarav.sharma@gmail.com",
   phone: "9876500001",
   city: "Lucknow",
   cgpa: 8.5 },
 {
   student_id: 2,
   name: "Ananya Verma",
   age: 19,
   gender: "Female",
   semester: 1,
   section: "A",
   email: "ananya.verma@gmail.com",
   phone: "9876500002",
   city: "Kanpur",
   cgpa: 8.7
 },
```

```
{
  student_id: 3,
  name: "Aditya Singh",
  age: 20,
  gender: "Male",
  semester: 1,
  section: "B",
  email: "aditya.singh@gmail.com",
  phone: "9876500003",
  city: "Lucknow",
  cgpa: 7.9
},
{
  student_id: 4,
  name: "Priya Gupta",
  age: 19,
  gender: "Female",
  semester: 1,
  section: "B",
  email: "priya.gupta@gmail.com",
  phone: "9876500004",
  city: "Agra",
  cgpa: 9.1
},
{
  student_id: 5,
  name: "Rohan Yadav",
```

```
  age: 20,
  gender: "Male",
  semester: 1,
  section: "A",
  email: "rohan.yadav@gmail.com",
  phone: "9876500005",
  city: "Ayodhya",
  cgpa: 7.5
},
{
  student_id: 6,
  name: "Sneha Mishra",
  age: 19,
  gender: "Female",
  semester: 1,
  section: "A",
  email: "sneha.mishra@gmail.com",
  phone: "9876500006",
  city: "Prayagraj",
  cgpa: 8.8
},
{
  student_id: 7,
  name: "Arjun Kumar",
  age: 20,
  gender: "Male",
  semester: 2,
```

```
  section: "B",
  email: "arjun.kumar@gmail.com",
  phone: "9876500007",
  city: "Varanasi",
  cgpa: 7.8
},
{
  student_id: 8,
  name: "Neha Singh",
  age: 19,
  gender: "Female",
  semester: 2,
  section: "A",
  email: "neha.singh@gmail.com",
  phone: "9876500008",
  city: "Lucknow",
  cgpa: 8.4
},
{
  student_id: 9,
  name: "Rahul Verma",
  age: 21,
  gender: "Male",
  semester: 2,
  section: "B",
  email: "rahul.verma@gmail.com",
  phone: "9876500009",
```

```
  city: "Noida",
  cgpa: 7.2
},
{
  student_id: 10,
  name: "Kavya Sharma",
  age: 19,
  gender: "Female",
  semester: 2,
  section: "A",
  email: "kavya.sharma@gmail.com",
  phone: "9876500010",
  city: "Kanpur",
  cgpa: 9.0 },
{
  student_id: 11,
  name: "Vivek Pandey",
  age: 20,
  gender: "Male",
  semester: 2,
  section: "B",
  email: "vivek.pandey@gmail.com",
  phone: "9876500011",
  city: "Lucknow",
  cgpa: 8.1  },
{
  student_id: 12,
```

```
  name: "Isha Gupta",
  age: 19,
  gender: "Female",
  semester: 2,
  section: "A",
  email: "isha.gupta@gmail.com",
  phone: "9876500012",
  city: "Agra",
  cgpa: 8.6  },
{
  student_id: 13,
  name: "Kunal Singh",
  age: 21,
  gender: "Male",
  semester: 3,
  section: "A",
  email: "kunal.singh@gmail.com",
  phone: "9876500013",
  city: "Meerut",
  cgpa: 7.6  },
{
  student_id: 14,
  name: "Simran Khan",
  age: 20,
  gender: "Female",
  semester: 3,
  section: "B",
```

```
  email: "simran.khan@gmail.com",
  phone: "9876500014",
  city: "Lucknow",
  cgpa: 8.9  },
{
  student_id: 15,
  name: "Yash Raj",
  age: 20,
  gender: "Male",
  semester: 3,
  section: "A",
  email: "yash.raj@gmail.com",
  phone: "9876500015",
  city: "Patna",
  cgpa: 7.4  },
{
  student_id: 16,
  name: "Pooja Verma",
  age: 19,
  gender: "Female",
  semester: 3,
  section: "B",
  email: "pooja.verma@gmail.com",
  phone: "9876500016",
  city: "Kanpur",
  cgpa: 8.3  },
{
```

```
  student_id: 17,
  name: "Mohit Sharma",
  age: 21,
  gender: "Male",
  semester: 3,
  section: "A",
  email: "mohit.sharma@gmail.com",
  phone: "9876500017",
  city: "Delhi",
  cgpa: 7.9  },
{
  student_id: 18,
  name: "Riya Gupta",
  age: 20,
  gender: "Female",
  semester: 3,
  section: "B",
  email: "riya.gupta@gmail.com",
  phone: "9876500018",
  city: "Noida",
  cgpa: 9.2  },
{
  student_id: 19,
  name: "Aman Tiwari",
  age: 21,
  gender: "Male",
  semester: 3,
```

```
  section: "A",
  email: "aman.tiwari@gmail.com",
  phone: "9876500019",
  city: "Lucknow",
  cgpa: 8.0 },
{
  student_id: 20,
  name: "Shreya Singh",
  age: 20,
  gender: "Female",
  semester: 3,
  section: "B",
  email: "shreya.singh@gmail.com",
  phone: "9876500020",
  city: "Varanasi",
  cgpa: 8.5  },
{
  student_id: 21,
  name: "Dev Kumar",
  age: 21,
  gender: "Male",
  semester: 4,
  section: "A",
  email: "dev.kumar@gmail.com",
  phone: "9876500021",
  city: "Lucknow",
  cgpa: 7.7  },
```

```
{
  student_id: 22,
  name: "Muskan Sharma",
  age: 20,
  gender: "Female",
  semester: 4,
  section: "B",
  email: "muskan.sharma@gmail.com",
  phone: "9876500022",
  city: "Agra",
  cgpa: 8.5  },
{
  student_id: 23,
  name: "Harsh Verma",
  age: 21,
  gender: "Male",
  semester: 4,
  section: "A",
  email: "harsh.verma@gmail.com",
  phone: "9876500023",
  city: "Kanpur",
  cgpa: 8.2  },
{
  student_id: 24,
  name: "Nandini Singh",
  age: 20,
  gender: "Female",
```

```
  semester: 4,
  section: "B",
  email: "nandini.singh@gmail.com",
  phone: "9876500024",
  city: "Lucknow",
  cgpa: 9.1  },
{
  student_id: 25,
  name: "Akash Gupta",
  age: 21,
  gender: "Male",
  semester: 4,
  section: "A",
  email: "akash.gupta@gmail.com",
  phone: "9876500025",
  city: "Prayagraj",
  cgpa: 7.8  },
{
  student_id: 26,
  name: "Sakshi Yadav",
  age: 20,
  gender: "Female",
  semester: 4,
  section: "B",
  email: "sakshi.yadav@gmail.com",
  phone: "9876500026",
  city: "Ayodhya",
```

```
  cgpa: 8.7 },
{
  student_id: 27,
  name: "Rajat Mishra",
  age: 21,
  gender: "Male",
  semester: 4,
  section: "A",
  email: "rajat.mishra@gmail.com",
  phone: "9876500027",
  city: "Varanasi",
  cgpa: 7.3  },
{
  student_id: 28,
  name: "Aditi Sharma",
  age: 20,
  gender: "Female",
  semester: 4,
  section: "B",
  email: "aditi.sharma@gmail.com",
  phone: "9876500028",
  city: "Delhi",
  cgpa: 9.0  },
{
  student_id: 29,
  name: "Nikhil Singh",
  age: 21,
```

```
  gender: "Male",
  semester: 4,
  section: "A",
  email: "nikhil.singh@gmail.com",
  phone: "9876500029",
  city: "Noida",
  cgpa: 8.1  },
{
  student_id: 30,
  name: "Tanya Verma",
  age: 20,
  gender: "Female",
  semester: 4,
  section: "B",
  email: "tanya.verma@gmail.com",
  phone: "9876500030",
  city: "Lucknow",
  cgpa: 8.9  },

{
  student_id: 31,
  name: "Abhishek Kumar",
  age: 21,
  gender: "Male",
  semester: 5,
  section: "A",
  email: "abhishek.kumar@gmail.com",
```

```
  phone: "9876500031",
  city: "Kanpur",
  cgpa: 7.5 },
{
  student_id: 32,
  name: "Shivani Gupta",
  age: 21,
  gender: "Female",
  semester: 5,
  section: "B",
  email: "shivani.gupta@gmail.com",
  phone: "9876500032",
  city: "Agra",
  cgpa: 8.6  },
{
  student_id: 33,
  name: "Manish Yadav",
  age: 22,
  gender: "Male",
  semester: 5,
  section: "A",
  email: "manish.yadav@gmail.com",
  phone: "9876500033",
  city: "Lucknow",
  cgpa: 7.9  },
{
  student_id: 34,
```

```
  name: "Palak Singh",
  age: 21,
  gender: "Female",
  semester: 5,
  section: "B",
  email: "palak.singh@gmail.com",
  phone: "9876500034",
  city: "Varanasi",
  cgpa: 9.3  },
{
  student_id: 35,
  name: "Saurabh Sharma",
  age: 22,
  gender: "Male",
  semester: 5,
  section: "A",
  email: "saurabh.sharma@gmail.com",
  phone: "9876500035",
  city: "Prayagraj",
  cgpa: 8.0 },
{
  student_id: 36,
  name: "Komal Verma",
  age: 21,
  gender: "Female",
  semester: 5,
  section: "B",
```

```
  email: "komal.verma@gmail.com",
  phone: "9876500036",
  city: "Kanpur",
  cgpa: 8.4  },
{
  student_id: 37,
  name: "Vishal Kumar",
  age: 22,
  gender: "Male",
  semester: 5,
  section: "A",
  email: "vishal.kumar@gmail.com",
  phone: "9876500037",
  city: "Lucknow",
  cgpa: 7.7  },
{
  student_id: 38,
  name: "Divya Mishra",
  age: 21,
  gender: "Female",
  semester: 5,
  section: "B",
  email: "divya.mishra@gmail.com",
  phone: "9876500038",
  city: "Ayodhya",
  cgpa: 8.9  },
{
```

```
  student_id: 39,
  name: "Rishabh Gupta",
  age: 22,
  gender: "Male",
  semester: 5,
  section: "A",
  email: "rishabh.gupta@gmail.com",
  phone: "9876500039",
  city: "Noida",
  cgpa: 8.3  },
{
  student_id: 40,
  name: "Mansi Sharma",
  age: 21,
  gender: "Female",
  semester: 5,
  section: "B",
  email: "mansi.sharma@gmail.com",
  phone: "9876500040",
  city: "Delhi",
  cgpa: 9.0  },

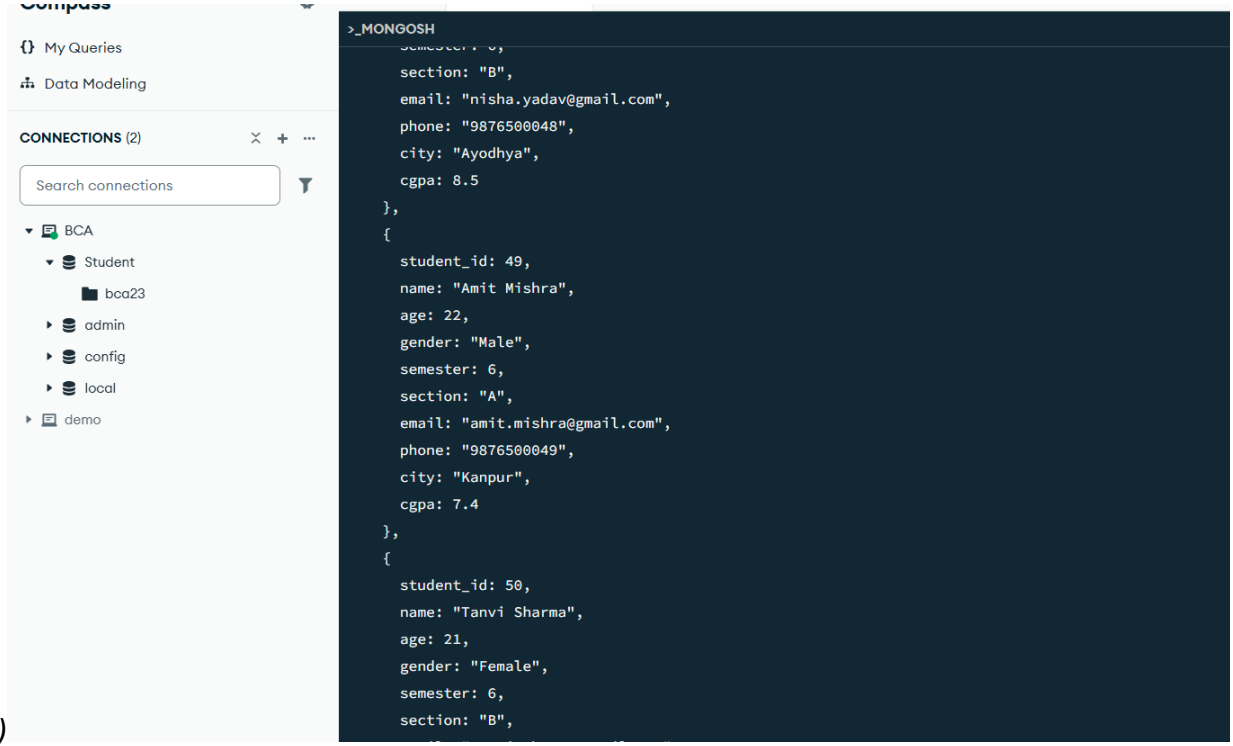
{
  student_id: 41,
  name: "Varun Singh",
  age: 22,
  gender: "Male",
```

```
  semester: 6,
  section: "A",
  email: "varun.singh@gmail.com",
  phone: "9876500041",
  city: "Lucknow",
  cgpa: 8.1  },
{
  student_id: 42,
  name: "Ritika Gupta",
  age: 21,
  gender: "Female",
  semester: 6,
  section: "B",
  email: "ritika.gupta@gmail.com",
  phone: "9876500042",
  city: "Agra",
  cgpa: 8.8  },
{
  student_id: 43,
  name: "Siddharth Kumar",
  age: 22,
  gender: "Male",
  semester: 6,
  section: "A",
  email: "siddharth.kumar@gmail.com",
  phone: "9876500043",
  city: "Kanpur",
```

```
  cgpa: 7.6  },
{
  student_id: 44,
  name: "Pallavi Singh",
  age: 21,
  gender: "Female",
  semester: 6,
  section: "B",
  email: "pallavi.singh@gmail.com",
  phone: "9876500044",
  city: "Varanasi",
  cgpa: 9.2  },
{
  student_id: 45,
  name: "Ravi Sharma",
  age: 22,
  gender: "Male",
  semester: 6,
  section: "A",
  email: "ravi.sharma@gmail.com",
  phone: "9876500045",
  city: "Lucknow",
  cgpa: 8.0  },
{
  student_id: 46,
  name: "Ayesha Khan",
  age: 21,
```

```
  gender: "Female",
  semester: 6,
  section: "B",
  email: "ayesha.khan@gmail.com",
  phone: "9876500046",
  city: "Delhi",
  cgpa: 8.7  },
{
  student_id: 47,
  name: "Deepak Verma",
  age: 22,
  gender: "Male",
  semester: 6,
  section: "A",
  email: "deepak.verma@gmail.com",
  phone: "9876500047",
  city: "Prayagraj",
  cgpa: 7.8  },
{
  student_id: 48,
  name: "Nisha Yadav",
  age: 21,
  gender: "Female",
  semester: 6,
  section: "B",
  email: "nisha.yadav@gmail.com",
  phone: "9876500048",
```

```
  city: "Ayodhya",
  cgpa: 8.5 },
{
  student_id: 49,
  name: "Amit Mishra",
  age: 22,
  gender: "Male",
  semester: 6,
  section: "A",
  email: "amit.mishra@gmail.com",
  phone: "9876500049",
  city: "Kanpur",
  cgpa: 7.4  },
{
  student_id: 50,
  name: "Tanvi Sharma",
  age: 21,
  gender: "Female",
  semester: 6,
  section: "B",
  email: "tanvi.sharma@gmail.com",
  phone: "9876500050",
  city: "Lucknow",
  cgpa: 9.1  }
```



])

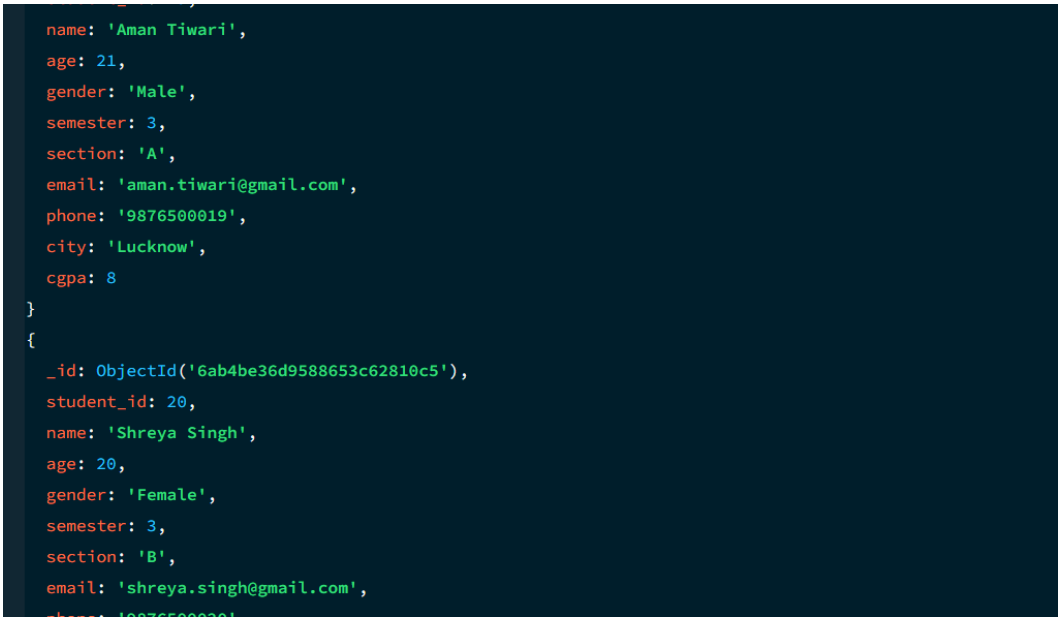
Explanation: Inserts multiple student documents into the collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

db. bca23. Find ()

Explanation: Retrieves documents from the students collection.

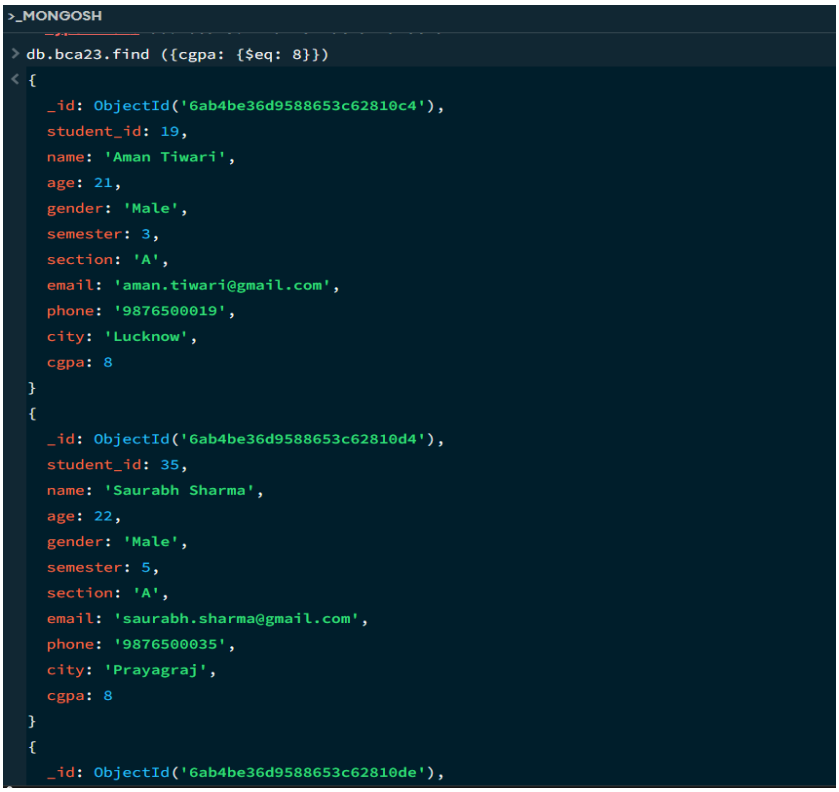


5. EQUAL TO — $eq

MongoDB Query:

db. bca23. Find ({cgpa: {$eq: 8}})

Explanation: Finds students whose cgpa are exactly 8.

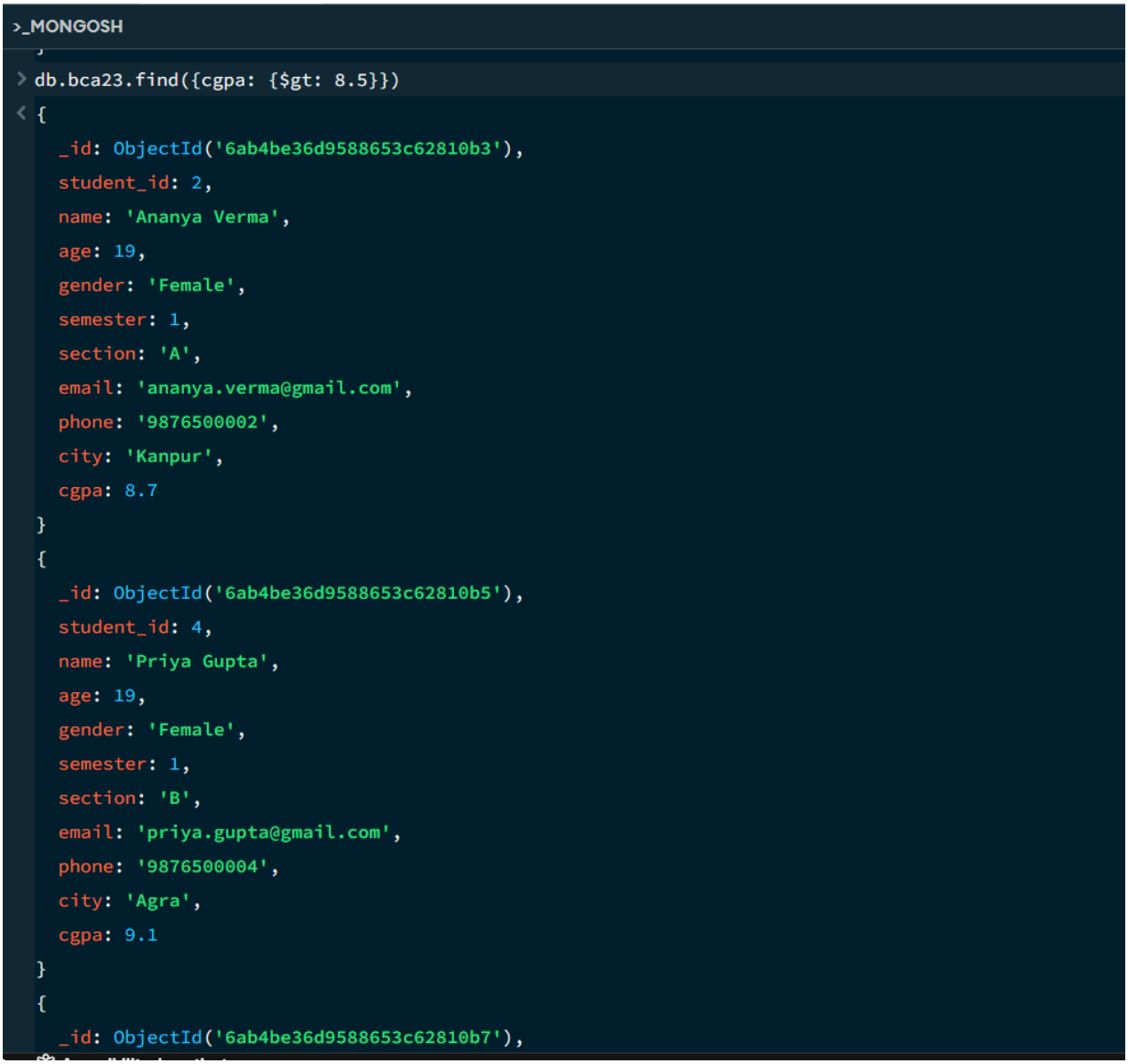


6. GREATER THAN — $gt

MongoDB Query:

db.bca23.find({cgpa: {$gt: 8.5}})

Explanation: Finds students whose cgpa are greater than 8.5.



7. LESS THAN — $lt

MongoDB Query:

db.students.find ({age: {$lt: 20}})

Explanation: Finds students whose age are less than 20.

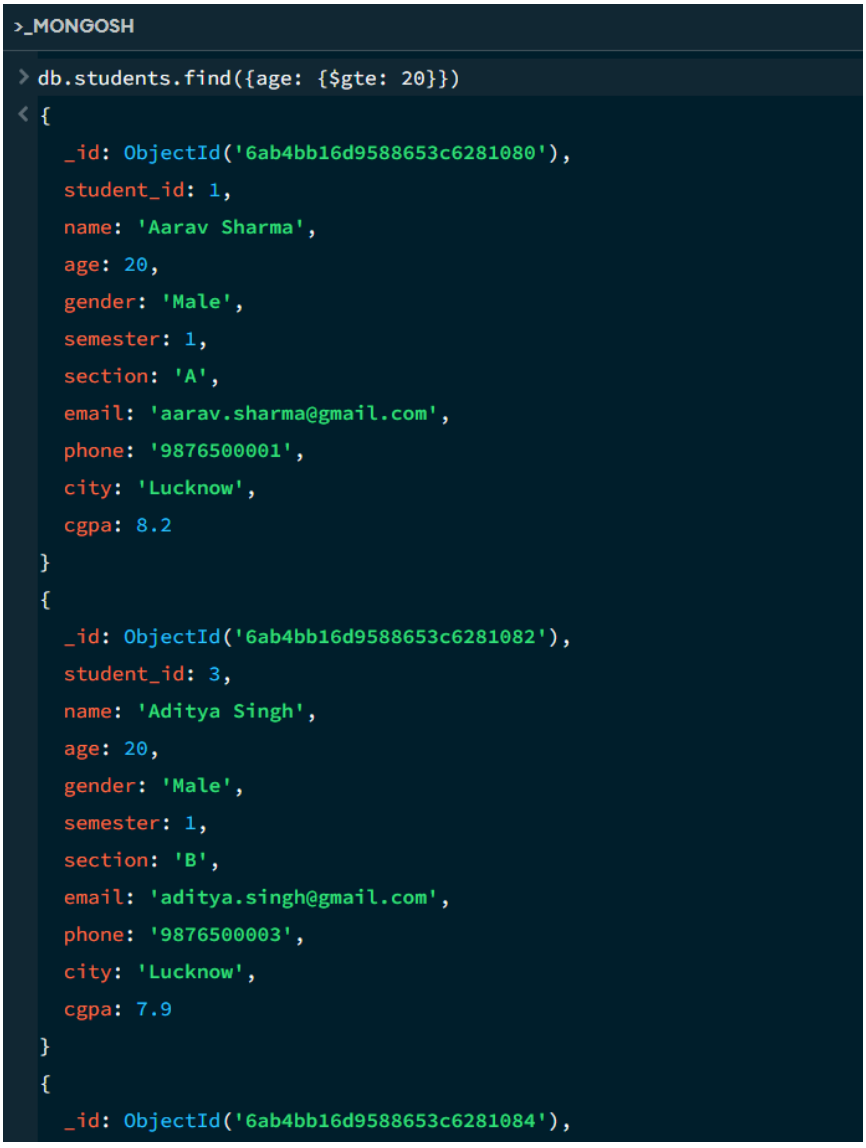


8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

db. students. find({age: {$gte: 20}})

Explanation: Finds students whose age are 20 or above.



9. LESS THAN OR EQUAL TO — $lte

MongoDB Query:

db. students. find({cgpa: {$lte: 8.5}})

Explanation: Finds students whose cgpa are 8.5 or below.



10. UPDATE OPERATION

MongoDB Query:

```
db.bca23.updateOne(
  {student_id:5},
  {$set:{cgpa:9.5}}
)
```

Explanation: Updates the cgpa to 9.5 of the student with student_id: 5.

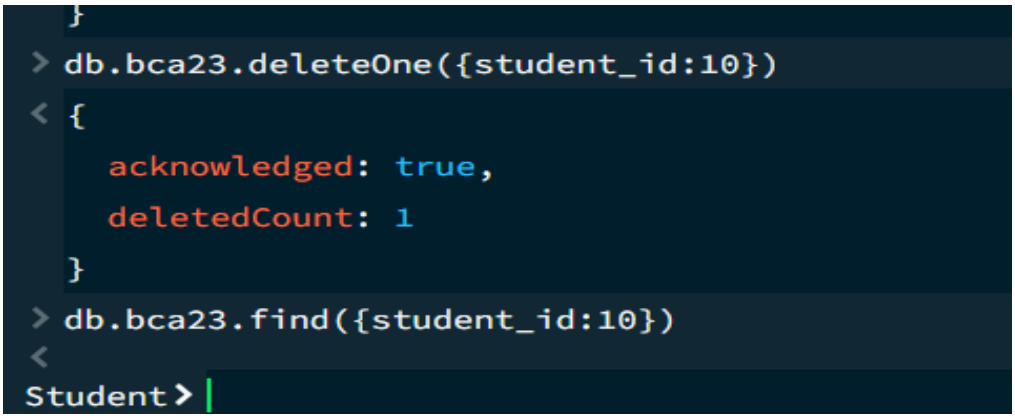
```
>_MONGOSH
    cgpa: 7.9
  }
  Type "it" for more
> db.bca23.updateOne(
    {student_id:5},
    {$set:{cgpa:9.5}}
  )
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
> db.bca23.find({student_id:5})
< {
    _id: ObjectId('6ab4be36d9588653c62810b6'),
    student_id: 5,
    name: 'Rohan Yadav',
    age: 20,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'rohan.yadav@gmail.com',
    phone: '9876500005',
    city: 'Ayodhya',
    cgpa: 9.5
  }
Student>
```

11. DELETE ONE OPERATION

MongoDB Query:

db.bca23.deleteOne({student_id:10})

Explanation: Deletes one document matching student_id: 10.



12. DELETE MANY OPERATION

MongoDB Query:

```
db.bca23.deleteMany({
  cgpa: {$lt:7.5}
})
```

Explanation: Deletes all students having cgpa below 7.5.

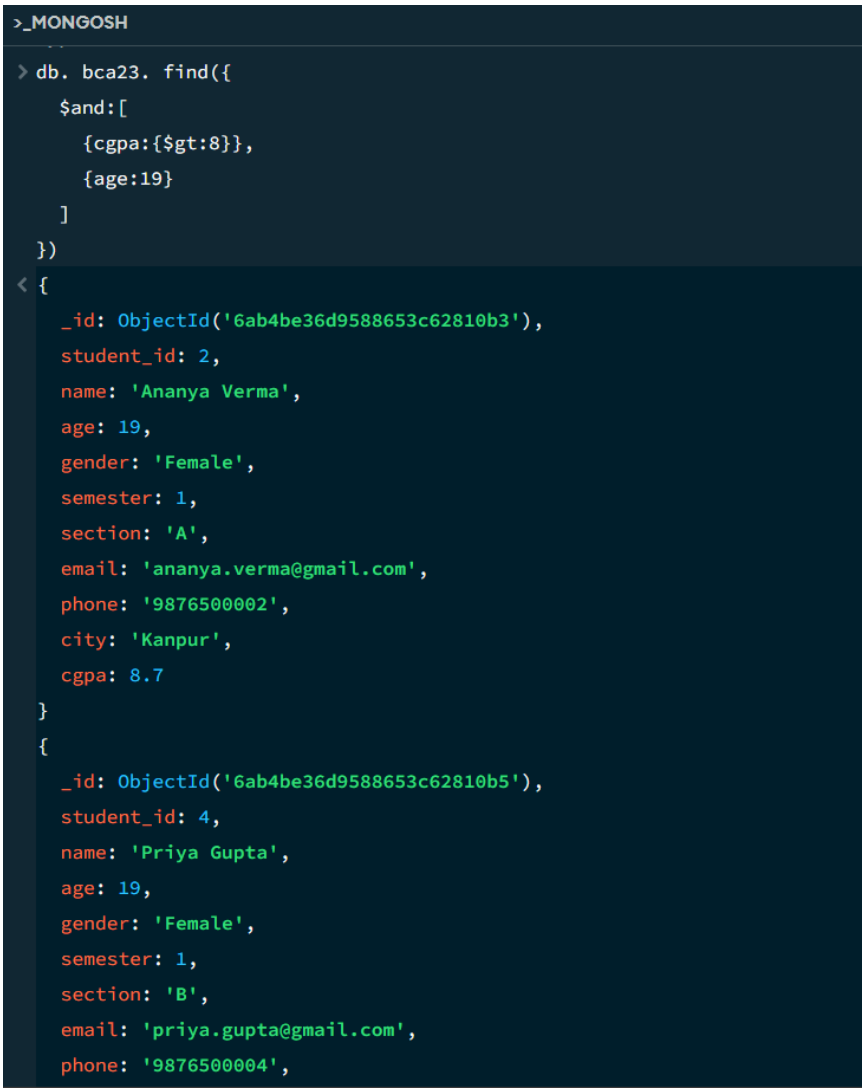
```
> db.bca23.find({student_id:10})
<
> db.bca23.deleteMany({
    cgpa: {$lt:7.5}
  })
< {
    acknowledged: true,
    deletedCount: 4
  }
> db.bca23.find({cgpa:7.5})
< {
    _id: ObjectId('6ab4be36d9588653c62810d0'),
    student_id: 31,
    name: 'Abhishek Kumar',
    age: 21,
    gender: 'Male',
    semester: 5,
    section: 'A',
    email: 'abhishek.kumar@gmail.com',
    phone: '9876500031',
    city: 'Kanpur',
    cgpa: 7.5
  }
> db.bca23.find({cgpa:7})
<
Student>
```

13. AND OPERATION — $and

MongoDB Query:

```
db. bca23. find({
  $and:[
    {cgpa:{$gt:8}},
    {age:19}
  ]
})
```

Explanation: Finds bca23 satisfying both conditions.

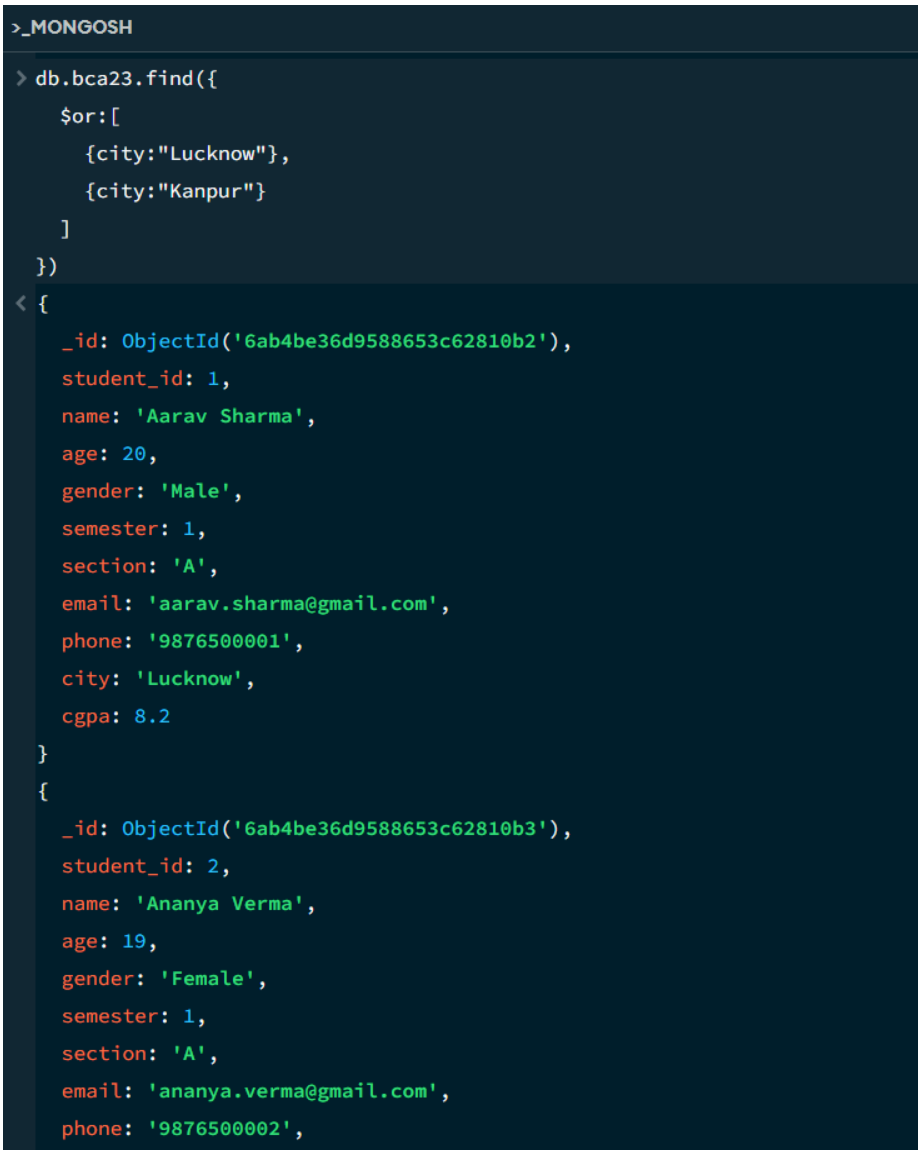


14. OR OPERATION — $or

MongoDB Query:

```
db.bca23.find({
  $or:[
    {city:"Lucknow"},
    {city:"Kanpur"}
  ]
})
```

Explanation: Finds bca23 from Lucknow or Kanpur.



15. SORT OPERATION

A.MongoDB Query:

db.bca23.find (). sort({age : 1})

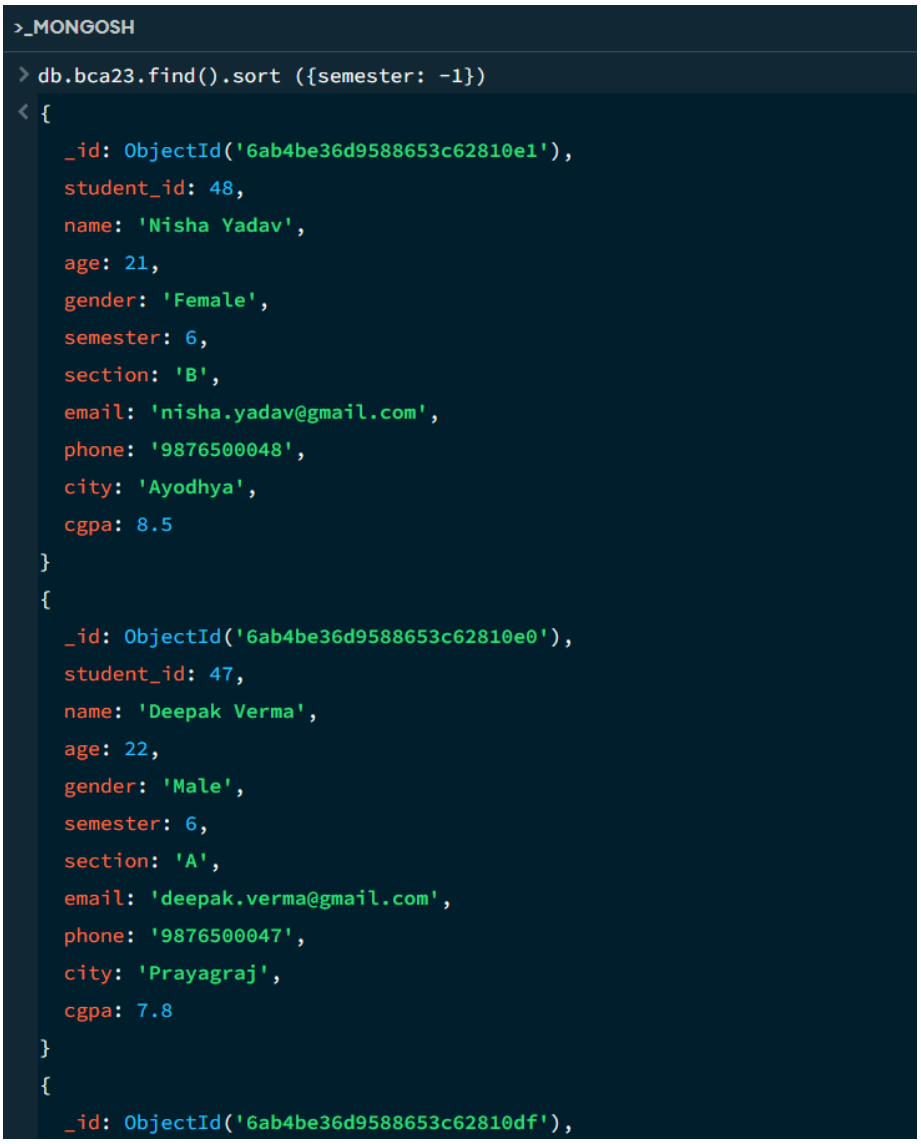
Explanation: Sorts students by age . Use 1 for ascending order.



B. MongoDB Query:

db.bca23.find().sort ({semester: -1})

Explanation: Sorts students by semester . Use -1 for descending order.



16. LIMIT OPERATION

MongoDB Query:

db.bca23.find().limit (5)

Explanation: Displays only the first five documents.

```
>_MONGOSH
  }
> db.bca23.find().limit (5)
< {
    _id: ObjectId('6ab4be36d9588653c62810b2'),
    student_id: 1,
    name: 'Aarav Sharma',
    age: 20,
    gender: 'Male',
    semester: 1,
    section: 'A',
    email: 'aarav.sharma@gmail.com',
    phone: '9876500001',
    city: 'Lucknow',
    cgpa: 8.2
  }
  {
    _id: ObjectId('6ab4be36d9588653c62810b3'),
    student_id: 2,
    name: 'Ananya Verma',
    age: 19,
    gender: 'Female',
    semester: 1,
    section: 'A',
    email: 'ananya.verma@gmail.com',
    phone: '9876500002',
    city: 'Kanpur',
    cgpa: 8.7
  }
  {
```

17. SEARCH BY GENDER

MongoDB Query:

db.bca23.find({gender:"Male"})

Explanation: Finds all students whose gender is male.

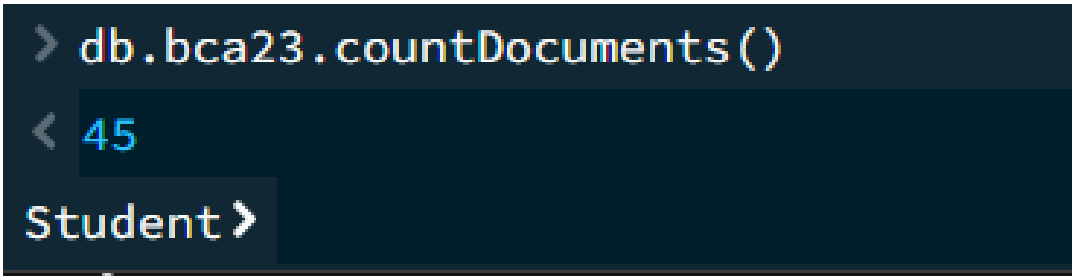


18. COUNT DOCUMENTS

MongoDB Query:

Db.bca23.countDocuments()

Explanation: Counts documents currently present in the collection (performing after the delete function).



MONGODB OPERATORS USED:

Operator	Meaning	Example
$eq	Equal to	cgpa: {$eq:8}
$gt	Greater than	cgpa: {$gt:85}
$lt	Less than	cgpa: {$lt:60}
$gte	Greater than or equal to	cgpa: {$gte:90}
$lte	Less than or equal to	cgpa: {$lte:70}
$and	All conditions true	$and:[...]
$or	At least one condition true	$or:[...]

CONCLUSION

This project provided practical knowledge of NoSQL databases using MongoDB. A Student Management System containing student information was created and different MongoDB operations were performed on the data.

Through this project, I learned how to insert, retrieve, update, and delete documents. I also learned how to Ascending order & descending order data, comparison operators such as $eq, $gt, $lt, $gte, and $lte work, along with logical operators such as $and and $or.

Overall, this project helped me understand how MongoDB can be used to manage and query student data efficiently.